Evolution of Microcystins Across Cyanobacteria

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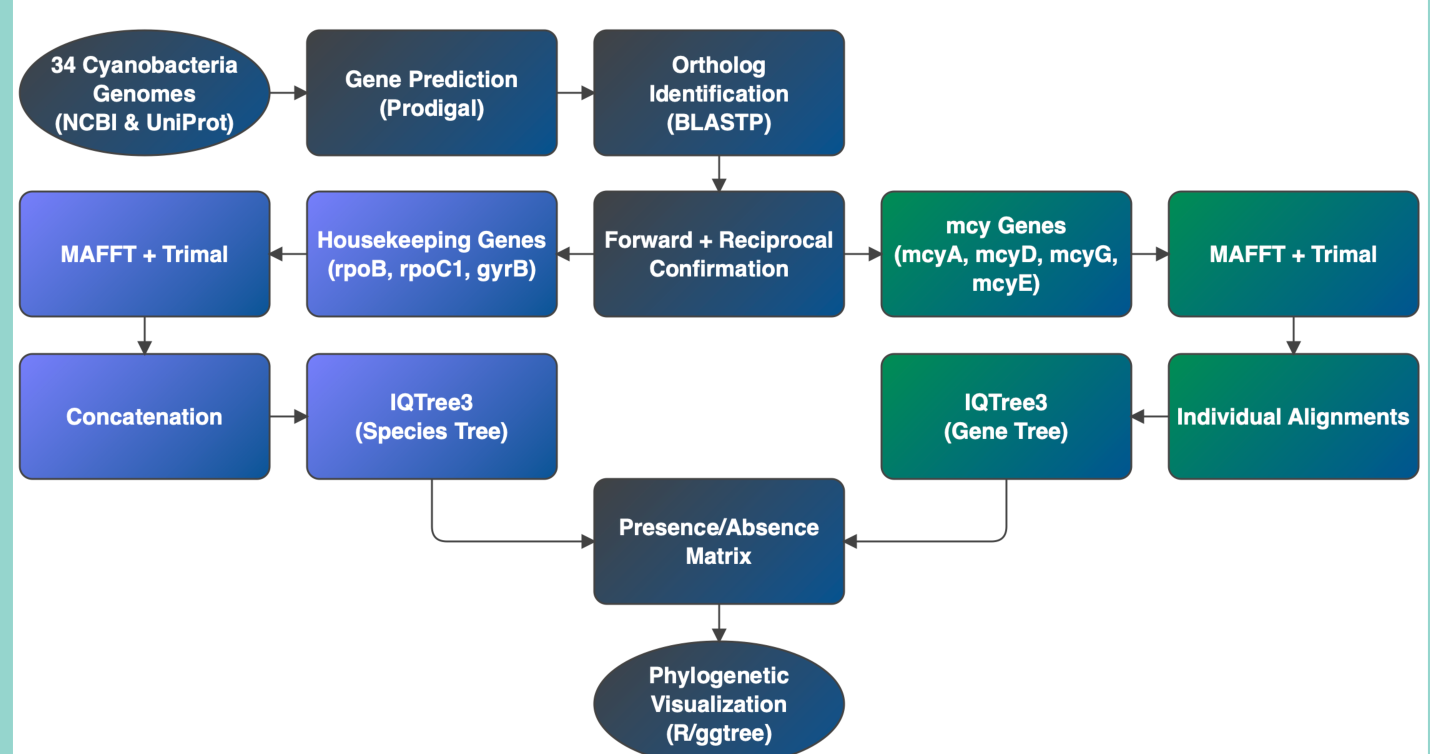
BIOL 4550 Molecular Evolution & Bioinformatics

Introduction

- Cyanobacteria is a large contributor to atmospheric oxygen and the source of harmful algal blooms (HABs) [6].
- Microcystins (mcy) are a hepatotoxin and are the dominant toxins found in HABs and present a threat to water quality and human health [1].
- Detection methods of microcystins are available but need refining [7] [9].
- Phylogenetics can optimize detection methods by determining conserved toxin producing genes to target with PCR.
- Microcystins are synthesized on large, homologous polyketidesynthase (PKS) and nonribosomal peptide synthetase (NRPS) enzyme complexes [5].
- This research poses two questions: (I) Does microcystin evolution follow cyanobacteria evolution? (II) Will using 4 mcy genes provide enough specificity for accurate molecular detection of microcystins.

Methodology



- 34 cyanobacteria genomes collected from NCBI representing diverse lineages.
- Protein coding sequences predicted using Prodigal.
- Forward BLASTP performed against all 34 genomes.
- Hits filtered by e-value ≤ 1e-5, percent identity ≥ 30%, & query coverage ≥ 70%.
- Reciprocal BLASTP confirmed true orthologs.
- False positives from homologous PKS/NRPS genes excluded by requiring all four mcy genes present.
- Species tree built from three conserved, housekeeping genes rpoB, rpoC1, & gyrB [4].
- Gene trees built from four mcy genes mcyA, mcyD, mcyE, & mcyG [8] [9] [10].
- Sequences aligned with MAFFT and trimmed with Trimal.
- Maximum likelihood trees inferred using IQTREE3 with 1000 ultrafast bootstraps (Species: LG+R4. mcyA, mcyD, mcyG: LG+F+I+G4, mcyE: JTTDCMUT+F+G4).
- Species tree rooted on Gloeobacter and gene trees rooted on midpoint due to absence of clear outgroup.
- Microcystis aeruginosa PCC 7806 selected as reference strain for mcy genes. Gloeobacter violaceus PCC 7421 selected as outgroup.

Results

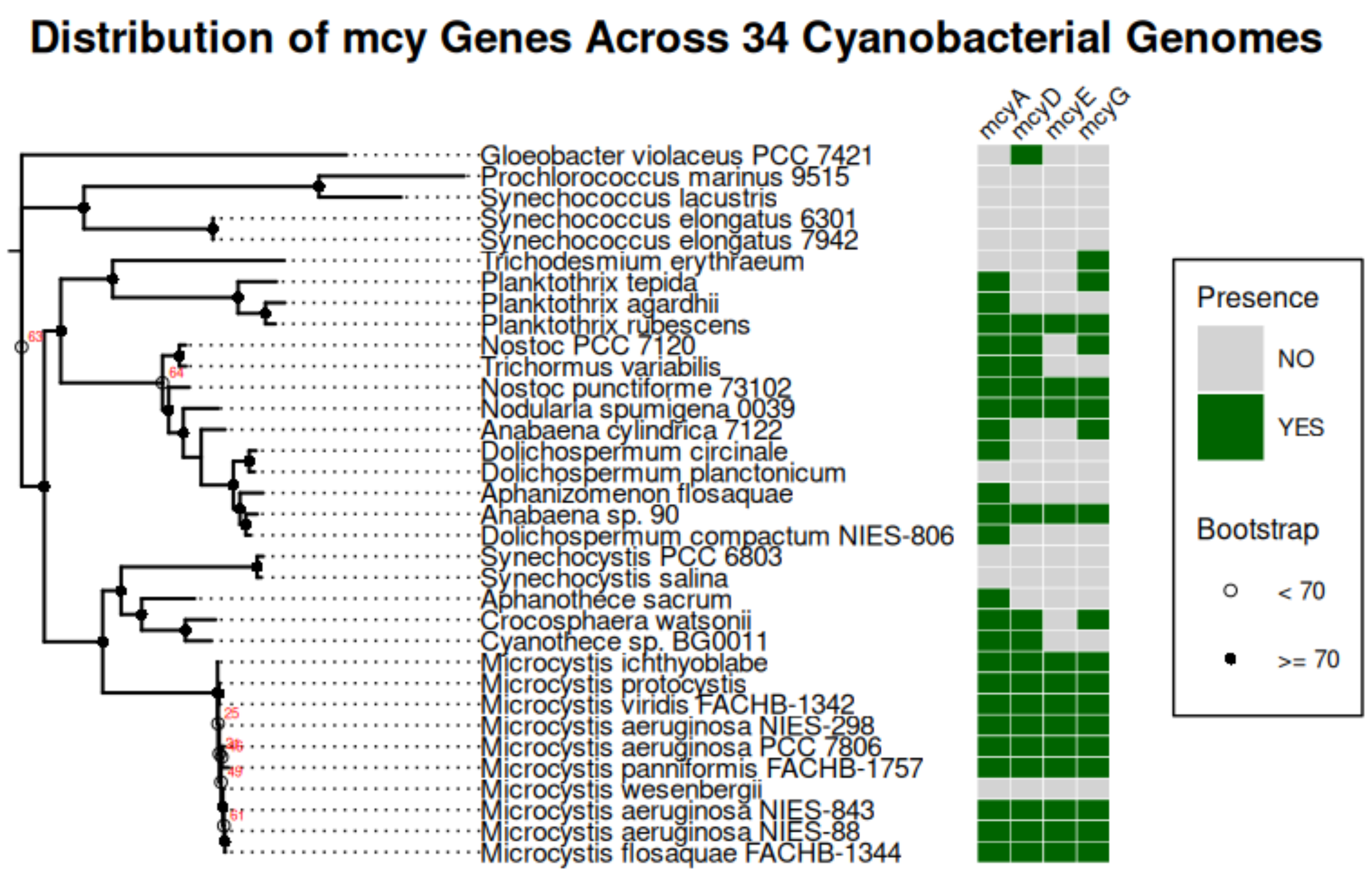


Figure 1. Topology of cyanobacteria genomes varying from known microcystin producers to non-producers. Red and open-circle bootstrap values indicate low confidence in node placement. (Right) Heatmap of orthologs detected for mcyA, mcyD, mcyE, and mcyG for each genome. A genome with all 4 mcy genes present indicate microcystin presence.

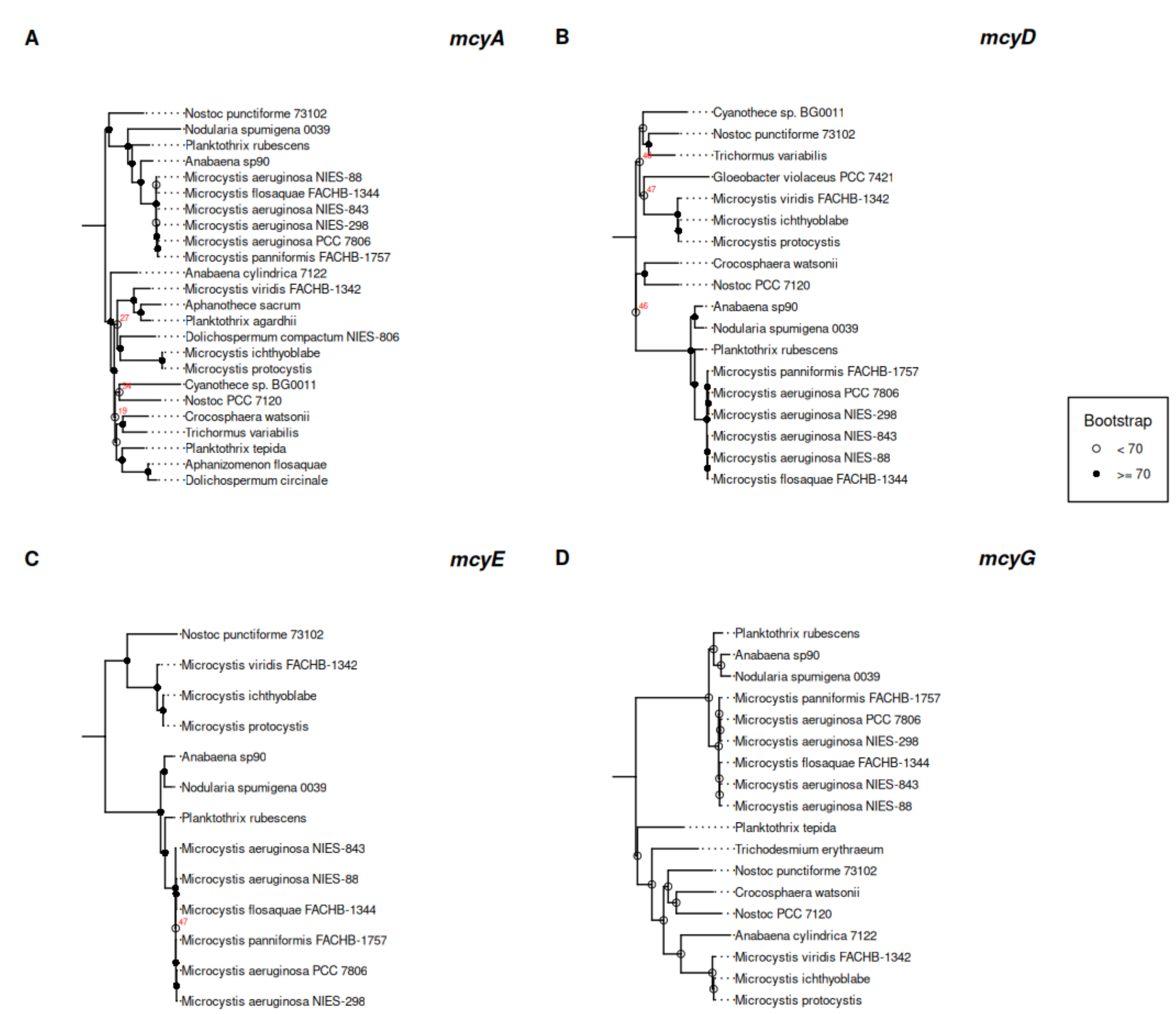


Figure 2. Individual gene phylogenies for each genome with a confirmed ortholog for each mcy gene.

Results cont.

Table 1. Table of potential false positive and false negative results. P. agardhii & D. circinale are known microcystin producers. N. punctiforme 73102 is a known non-producer. N. spumigena 0039 produces a homologous anatoxin.

Genome	Expected Microcystin Detection	Result	Assessment
Planktothrix agardhii	YES	Partial (mcyA only)	Likely false negative[2].
Dolichospermum circinale	YES	Partial (mcyA only)	Likely false negative.
Nostoc punctiforme 73102	NO	YES	Likely false positive.
Microcystis wesenbergii	Variable	NO	Possible true negative[7].
Nodularia spumigena 0039	Related	YES	Nodularin cluster homology.

Conclusions & Discussion

- Consistent clustering of Microcystis strains between the species and gene trees is consistent with vertical inheritance.
- Conflicting clustering of Microcystis and Planktothrix, Anabaena, & Nodularia between species and gene trees suggest horizontal gene transfer of mcy genes, though studies indicate an ancient ancestor where the genes were lost and reappeared [3].
- Positive ortholog of mcyD in G. violaceus evidences shared ancient PKS gene that can obscure microcystin detection.
- Likely false positive in N. punctiforme showcases using 4 mcy genes isn't specific enough for molecular detection of microcystins.
- Consistency within Microcystis strains and the Planktothrix,Anabaena, & Nodularia strains determine microcystin evolutionfollows cyanobacterial evolution within, but not across, genera.
- The selected 4 mcy genes does not provide enough specificity to improve molecular detection methods of microcystins.

References

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